

Face Occlusion Estimation via ConvNeXt Ensemble with Classical CV Baseline

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Introduction

The Telecom Paris–Idemia Face Occlusion Challenge frames face occlusion estimation as a regression task: given a 224×224 pixel aligned face crop, predict the fraction of the face occluded, $\hat{y} \in [0, 1]$. The official evaluation metric is a **gender-balanced, occlusion-weighted mean squared error**:

$$\mathcal{L} = \frac{E_F + E_M}{2} + |E_F - E_M|$$

where for each gender group $g \in \{F, M\}$:

$$E_g = \frac{\sum_i w_i (\hat{y}_i - y_i)^2}{\sum_i w_i}, \quad w_i = \frac{1}{30} + y_i$$

The weight w_i makes high-occlusion samples up to **31× more influential** than clear faces, heavily penalising false negatives on strongly occluded faces. The term $|E_F - E_M|$ further requires equitable performance across genders. The training set contains 100 000 images, the test set 29 980. The occlusion distribution is highly skewed (mean 0.083, 69% of samples have $GT < 0.1$) and gender-imbalanced (32.4% female, 67.6% male). Critically, **test gender labels are hidden** (replaced by x), making per-gender post-hoc calibration impossible.

Classical CV Baseline

Prior to deep learning, we built a **fully training-free classical CV heuristic** to establish an analytical upper bound. The pipeline: (1) CLAHE normalisation on the Y channel of YCrCb; (2) primary skin mask from fixed chrominance ranges ($Cr \in [133, 173]$, $Cb \in [77, 127]$); (3) adaptive HSV secondary mask; (4) Laplacian texture filter with anatomy-aware exclusion zones (eyes, mouth); (5) occlusion fraction estimated from the non-skin pixel ratio inside a central circular ROI. This zero-training approach hit a hard ceiling of **0.023** on the challenge metric. Compression artefacts at 224×224 , variable lighting, and the semantic richness of occlusion types (scarves, hands, hair) were insurmountable for fixed-colour detectors. This validated the shift to deep learning.

Deep Learning Solution

Architecture

Each model is a **transfer-learning regressor**: a pretrained backbone from the `timm` library [1] with `num_classes=0`, followed by a custom regression head:

```
Linear( $d_{\text{feat}} \rightarrow 512$ ) → ReLU → Dropout(0.3)
→ Linear(512 → 128) → ReLU → Dropout(0.3)
→ Linear(128 → 1) → Sigmoid
```

The terminal Sigmoid constrains predictions to $[0, 1]$ without output clipping. We explored EfficientNet-B4 [2], ConvNeXt-Base/Large [3], ConvNeXtV2-Base [4], ViT-Base, EfficientNetV2-M and Swin-Base.

Loss Function

The training loss **replicates the evaluation metric exactly**: $\mathcal{L}_{\text{train}} = \frac{E_F + E_M}{2} + |E_F - E_M|$, with per-sample weights $w_i = \frac{1}{30} + y_i$ computed inside each forward pass. In the rare case of a mono-gender mini-batch, the loss falls back to global weighted MSE. This eliminates any optimisation gap that would arise from vanilla MSE or Huber.

Training

Data split: stratified 80/20 on gender × occlusion quintile (seed 42).

Weighted sampler: draw probability proportional to $s_i = (\frac{1}{30} + y_i) \times \frac{N}{2N_g} \times \text{male_boost}$, correcting both occlusion imbalance and gender skew. `male_boost = 2.5` (moderate) reduces $|E_F - E_M|$ without overfitting.

Optimiser: AdamW [5], weight decay 10^{-4} , cosine annealing over 30 epochs. Learning rate: 10^{-4} for ConvNeXt, 3×10^{-5} for ConvNeXtV2 (FCMAE backbone diverges at 10^{-4}).

Augmentation: random horizontal flip, ColorJitter (0.3/0.3/0.2), rotation $\pm 20^\circ$, Gaussian blur ($p = 0.4$). RandomErasing is excluded (it would simulate occlusion and leak label information).

Inference: Test-Time Augmentation (TTA) with 15 passes (1 clean + 14 augmented), averaged.

Final Ensemble (Score 0.00112)

After training 12+ models, the best leaderboard result combines **seven ConvNeXt models** in two tiers:

- **ENS4** — four strong models: ConvNeXtV2-Base (seed 123), ConvNeXt-Large (seeds 123 and 999), ConvNeXt-Base (seed 456).
- **s6** — three diverse models: ConvNeXt-Base (seeds 42, 123), EfficientNet-B4 (seed 123).
- **Final**: $0.55 \cdot \text{ENS4} + 0.45 \cdot \text{s6}$.

Model / Ensemble	Detail	Score
ConvNeXt-Large (s. 999)	val	0.0011
ConvNeXt-Large (s. 123)	val	0.0011
ConvNeXtV2-Base (s. 123)	val	0.0014
ConvNeXt-Base (s. 42/123)	val	0.0016
ConvNeXt-Base (s. 456)	val	0.0018
EfficientNet-B4 (s. 123)	val	0.0017
ENS3 (3 ConvNeXt)	leaderboard	0.00115
s6 (Base + B4)	leaderboard	0.00116
ENS4 (4 ConvNeXt)	leaderboard	0.00115
Final blend	leaderboard	0.00112

Table 1: Per-model validation scores and leaderboard results.

Key Findings

Validation scores can mislead. A synthetic-degradation recipe (artificial occlusions, aggressive tail weighting, GT-scaled male boost) achieved the best val score (0.00129) but the **worst** test score (0.00218) — a 70% regression versus the clean ensemble. Synthetic tricks overfit the validation split. We learned to trust the leaderboard over local validation.

Architecture family matters more than weight tuning. ViT-Base, Swin-Base and EfficientNetV2-M were all clearly weaker (val > 0.0018); adding ViT **degraded** the ensemble to 0.00121. ConvNeXt and ConvNeXtV2 form the dominant family. EfficientNet-B4 at low weight (15% via s6) contributes useful architectural decorrelation. Fine-tuning the 0.55/0.45 blend ratio produced no further gain — the plateau at 0.00112 can only be broken by new, genuinely diverse models.

Test gender is hidden. Since `test_students.csv` contains no gender column, per-gender post-hoc calibration is impossible. Global (α, β) scaling on validation gave only 0.9% gain, confirming that the clean models are already well-calibrated. Gender balancing is confined to the training phase (`male_boost` + weighted sampler).

Limitations

Computational cost. Seven deep networks, each trained 10–20 h on an RTX 3090, plus $\text{TTA} \times 15$ inference (1–2 h per model). Not suited for real-time deployment.

Ensemble complexity. ConvNeXt-Large has 200 M parameters; the full ensemble is heavy in memory and storage. Several hyperparameters require careful tuning (`male_boost`, per-backbone learning rate, blend ratio).

Validation reliability. The interim leaderboard reflects a subset of the test distribution; the final score on the held-out set may differ.

Conclusion

We presented a two-stage solution: a training-free classical CV baseline (ceiling 0.023) that confirmed the necessity of deep learning, followed by a seven-model ConvNeXt ensemble that achieves **0.00112** on the interim leaderboard. The key design principles — loss identical to the

metric, occlusion- and gender-weighted sampling, TTA, and diverse multi-seed multi-architecture ensembling — each directly target the structure of the evaluation criterion. Critical lessons learned: validation overfitting is severe with synthetic augmentation, backbone family dominates over weight optimisation, and architectural diversity (not just seed diversity) is the primary driver of ensemble gains.

Bibliography

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